

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

C04: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2

Annexure A

Comparative Analysis

Code of Conduct:

I N. Yugesh(192424329)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:_ N Yugesh

1. CNNs vs. Vision Transformers (ViTs) for Object Detection

Scenario: A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Selection of Studies / Comparison Parameters

Three parameters were selected because together they map directly onto the scenario's three binding constraints — rare/small objects (feature extraction and small-object performance), limited labelled data (data efficiency), and real-time embedded deployment (computational cost):

- Feature extraction capability for local vs. global context
- Performance with small or sparse objects
- Computational complexity and inference speed for embedded systems

Comparison & Differentiation

Parameter	CNNs	Vision Transformers (ViTs)
Feature extraction: local vs. global context	Hierarchical local receptive fields with weight sharing give a strong locality/translation-equivariance bias. Local edges and textures are captured in early layers; global context builds up gradually only as depth/receptive field increases (pooling, strided convolutions).	Self-attention relates every patch to every other patch from the very first layer, so global context (e.g., scene layout, relation between a cone and the road edge) is available immediately. No built-in locality bias — the model has to learn spatial structure from data.
Performance with small or sparse objects	Inductive bias helps generalisation with less data, but repeated pooling/downsampling can wash out very small objects (a distant traffic cone) unless multi-scale features (FPN, multi-scale anchors) are added.	Patch tokenisation (e.g., 16x16 patches) can dilute fine detail on small objects unless small patch sizes or hierarchical designs (Swin, PVT) are used; without strong pretraining, the lack of locality prior makes ViTs more data-hungry and prone to underperforming on small/rare classes with limited labels.
Computational complexity & inference speed (embedded)	Convolution cost scales roughly linearly with image size; well-supported by mature embedded toolchains (TensorRT, OpenVINO) and lightweight backbones (MobileNet, YOLO-tiny/-nano) built specifically for low-latency, low-memory inference.	Vanilla self-attention cost scales quadratically with the number of tokens (image size), making it memory- and compute-heavy at real-time resolutions; efficient variants (Swin's windowed attention, MobileViT, EfficientFormer) are needed to approach CNN-level latency on embedded hardware.

Critical Evaluation

- CNNs' strength — the built-in locality/weight-sharing prior — is exactly what limited labelled data rewards: the hypothesis space is constrained, so the network needs fewer examples per class to generalise, which matters when rare classes like "traffic cone" or "animal" may have only a few hundred labelled instances.
- CNNs' weakness is that pure convolutional stacks build global context slowly and can still miss context-dependent cases (e.g., an animal recognisable mainly from its relation to the road, not local texture alone) unless the receptive field is deliberately widened.
- ViTs' strength — immediate global self-attention — is valuable exactly where CNNs are weak: reasoning about an ambiguous, partially-occluded object using distant scene cues. This is a real advantage for rare, hard-to-classify obstacles.

- ViTs' weakness is a compounding one for this scenario: no locality bias plus limited labelled data plus a quadratic-cost attention mechanism together push against real-time embedded feasibility, unless the model is pretrained at scale and/or redesigned as a hierarchical/efficient ViT.
- Neither model is unconditionally better — the verdict depends on whether large-scale pretraining and an efficient ViT variant are available; without those, the CNN's inductive bias and mature embedded tooling make it the safer default for this exact scenario.

Synthesis & Insight — Real-Time Deployment and the Role of Transfer Learning / Augmentation

For real-time deployment on embedded hardware with limited labelled data, a CNN-based detector (or a hybrid CNN-ViT / hierarchical ViT such as Swin) is currently the more practical choice: it has lower latency and memory footprint, mature embedded deployment tooling, and its inductive bias generalises better from a small labelled set. Pure vanilla ViTs remain attractive for their global-context reasoning but are harder to justify under real-time and data constraints unless mitigations are applied.

1. Transfer learning: initialising from a backbone pretrained on a large dataset (ImageNet-21k, or a large driving dataset such as BDD100K/nuScenes) substantially closes the gap — a ViT pretrained at scale largely compensates for its missing locality bias, and a pretrained CNN backbone converges faster and generalises better on the rare-class, limited-label regime described in the scenario.
2. Data augmentation: because traffic cones and animals are rare classes, augmentation strategies such as copy-paste augmentation (pasting rare-object crops into diverse backgrounds), domain randomisation, and synthetic data generation directly increase the effective number of rare-class examples seen during training — this benefits both architectures, but is especially important for ViTs, which lack a built-in prior to fall back on when data is scarce.

Recommended path: a CNN or hierarchical-ViT backbone (e.g., Swin-Tiny / MobileViT) pretrained on a large driving-domain dataset, fine-tuned with rare-class copy-paste augmentation, and exported through an embedded-optimised runtime — balancing the real-time/limited-data constraints against the benefit of richer global context.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Selection of Studies / Comparison Parameters

The three parameters below were chosen because the scenario's defining constraint — varying topology and vertex density across meshes — directly stresses the assumptions each method relies on:

- Ability to handle non-rigid deformations
- Robustness to varying mesh resolutions and noise
- Ease of implementation and computational cost

Comparison & Differentiation

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Handling non-rigid deformations	Models deformation as a linear combination of eigenvector basis shapes learned from an aligned training set (e.g., 3D Morphable Models, Active Shape Models). Captures common global deformation modes (identity, expression) well when the target deformation lies inside the learned linear subspace, but struggles with deformations outside it (extreme expressions, topology change).	Represents the mesh as a graph and learns/optimises correspondence through local connectivity and geometric features (e.g., spectral / functional maps, graph neural networks). Non-linear and localised deformations are captured more naturally because inference is driven by local structure and message-passing rather than a fixed global linear basis.
Robustness to varying resolutions & noise	Requires meshes to already be in dense, consistent vertex-to-vertex correspondence (same topology/vertex count) to build the shape basis — poor robustness out-of-the-box to varying topology or vertex density; noise leaks into the dominant components since the model captures global variance.	Intrinsic/spectral representations (e.g., Laplace-Beltrami eigenbasis, functional maps) are largely topology- and parameterisation-agnostic, so they tolerate differing vertex density and moderate resampling better; still sensitive to strong noise or large non-isometric distortion, since functional maps assume near-isometry.
Ease of implementation & computational cost	Simple to implement once trained; fitting is a fast linear least-squares problem (well-understood tooling: Basel Face Model, FLAME). However, building the training basis requires a pre-registered dataset with consistent topology — an expensive upfront data-preparation step.	More complex to implement (eigen-decomposition / spectral solvers, functional-map regularisers, or training a GNN); computationally heavier for large/high-resolution meshes, though tooling (pyFM, deep functional maps) is maturing and inference is fast once a model is trained.

Critical Evaluation

- PCA-based priors are efficient and easy to deploy, but their core assumption — a fixed, consistent mesh topology across all samples — is directly violated by the scenario's stated variation in topology and vertex density; using them as-is would require an expensive, error-prone remeshing/registration step before the method even applies.
- PCA's linear subspace is also a genuine representational limitation: realistic facial morphing often needs localised, asymmetric, non-linear deformations (a single raised eyebrow, an asymmetric smile) that a small number of global eigenmodes will tend to blur or over-smooth.
- Graph-based / spectral methods handle the topology/vertex-density variation far more gracefully because they reason over the mesh's intrinsic geometry rather than a fixed vertex ordering, which is the property this scenario most directly needs.

- The trade-off is cost and complexity: graph-based correspondence is more expensive to implement and compute, and spectral methods still assume the surfaces are near-isometric, which can break down for extreme expression changes — so it is not a free upgrade, only a better fit for this specific constraint.

Synthesis & Insight — Which Approach Gives More Accurate Correspondences for Realistic Morphing

Given that the scenario explicitly states meshes vary in topology and vertex density, graph-based / spectral correspondence methods (functional maps, deep functional maps, or GNN-based correspondence) are better suited and would be expected to produce more accurate correspondences: they operate on intrinsic surface geometry, so they do not require the meshes to already share a common topology, and they can represent the non-linear, localised deformation patterns that a global linear PCA basis tends to smooth over.

PCA-based shape priors are not obsolete in this pipeline, however — once dense correspondence has been established (by the graph-based method), a PCA-style linear basis is still a compact and computationally cheap representation for the downstream morphing/blending step itself, since morphing between two already-corresponded meshes is a much better-conditioned linear problem than establishing the correspondence in the first place.

Recommended path: use graph-based / spectral (deep functional map) correspondence as the primary method to align the varying-topology face meshes, then optionally fit a PCA-based shape model on top of the now-consistent correspondences to drive efficient, controllable morphing — combining each method's strength at the stage where it is actually needed.